

HTTP SECURITY HEADER CHECKER – PROJECT SUMMARY

Overview

This project is a Python-based HTTP Security Header Checker developed to analyze a website's security headers and identify missing protections that could expose the application to common web attacks.

The tool sends an HTTP request to a target website, retrieves the response headers, and checks for the presence of important security headers used to improve web application security.

Features Added

- Fixed URL validation issues
 - Professional ASCII banner with colored output
 - Security score calculation
 - Detection of missing security headers
 - Display of header values
 - Browser User-Agent support
 - Detailed recommendations for missing headers
 - Exception handling for network and connection errors
 - Improved user experience
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Security Headers Checked

- Strict-Transport-Security (HSTS)
- Content-Security-Policy (CSP)
- X-Frame-Options
- X-Content-Type-Options
- Permissions-Policy
- Referrer-Policy
- Cross-Origin-Opener-Policy
- Cross-Origin-Resource-Policy
- Cross-Origin-Embedder-Policy

Technologies Used

- Python
- Requests Library
- Colorama

Example Output

[+] Strict-Transport-Security Found

[+] X-Frame-Options Found

[-] Content-Security-Policy Missing

Security Score: 78%

Security Rating: Good

Concepts Demonstrated

- Web Security Fundamentals
- HTTP Response Headers
- Security Misconfiguration Detection
- Python Automation
- Error Handling
- Cybersecurity Assessment Techniques

Learning Outcome

This project helps understand how HTTP security headers protect websites against threats such as:

- Cross-Site Scripting (XSS)
- Clickjacking
- MIME-Type Sniffing
- Information Leakage

- Cross-Origin Attacks
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Disclaimer

This project is intended for educational purposes and authorized security testing only. Always obtain proper permission before assessing any website or web application.

30 Days of Cybersecurity Automation - Day 05

Building practical cybersecurity tools through hands-on automation and real-world security assessments.

Author: Zeeshan Alam